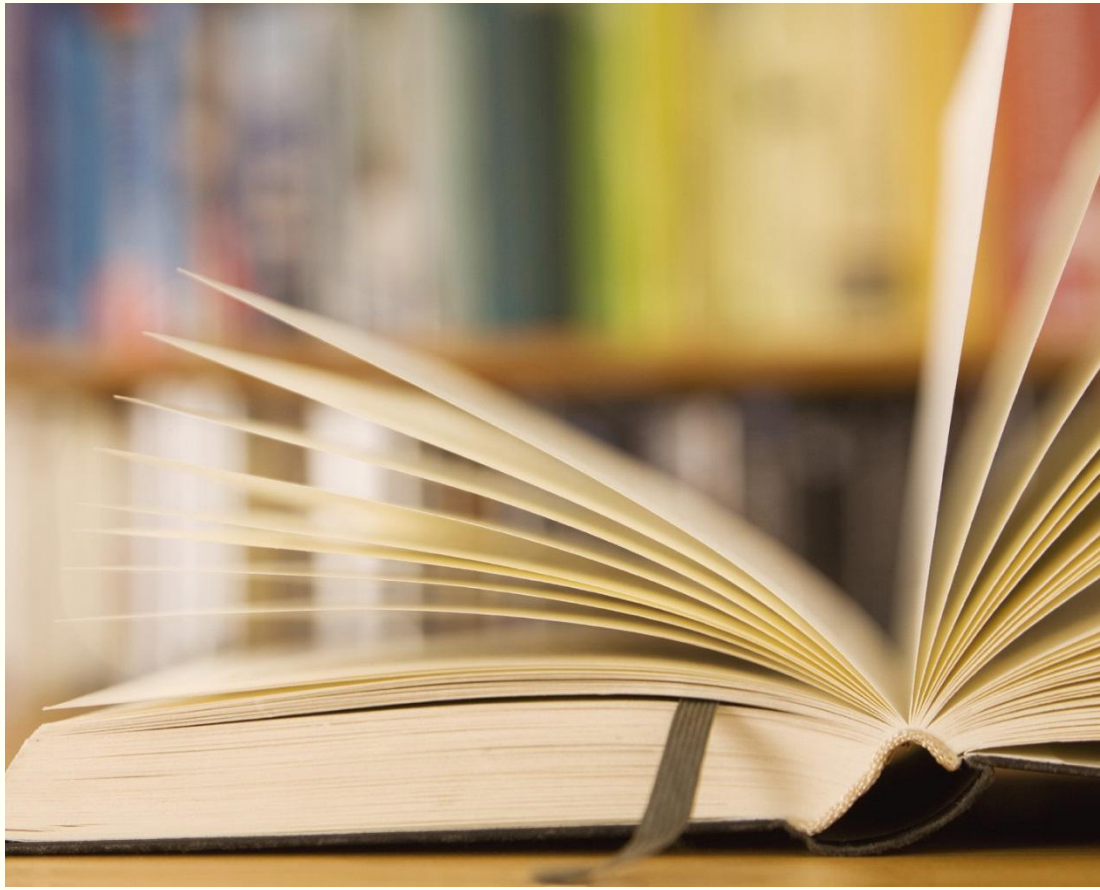
PHYSICAL LAYER: TRANSMISSION MEDIA



Unit II - Topics

- **Physical Layer**: Transmission Media - Guided Transmission Media – Twisted pair, Coaxial Cable, Fiber Optics, Unguided Media–Radio waves, Microwaves, Infrared.
- **Switching:** Packet and Circuit Switching
- **Data Link Layer:** Framing, Flow and Error Control, Error Detection and Correction - Types of Errors, Redundancy, LRC, CRC, Checksum. Flow and Error control Protocols: Stop and Wait, Stop and Wait ARQ, Go back-N ARQ, Selective repeat ARQ.

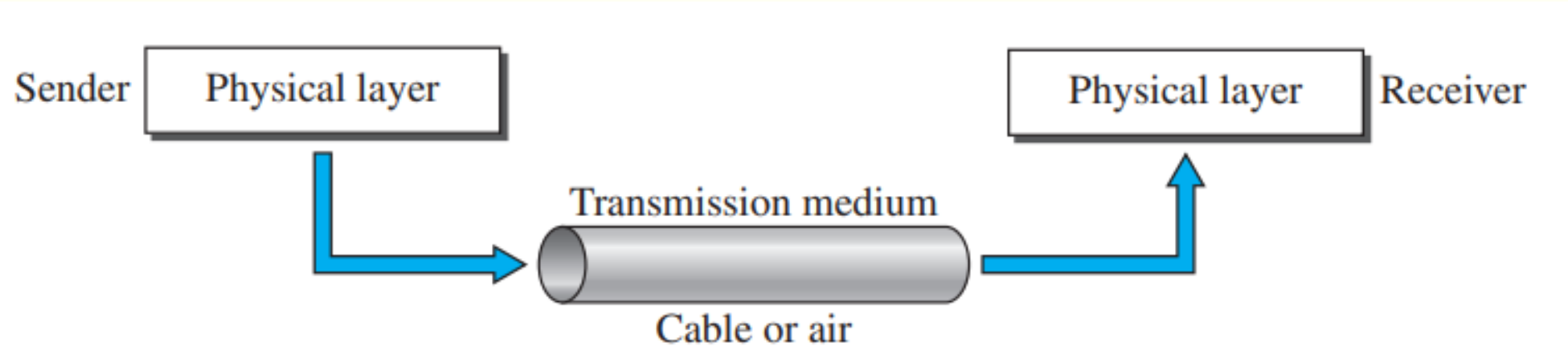
Contents

Transmission Media

- Guided Transmission Media – Twisted pair, Coaxial Cable, Fiber Optics
- Unguided Media – Radio waves, Microwaves, Infrared.

Transmission Media

- Transmission media are actually located below the physical layer and are directly controlled by the physical layer.

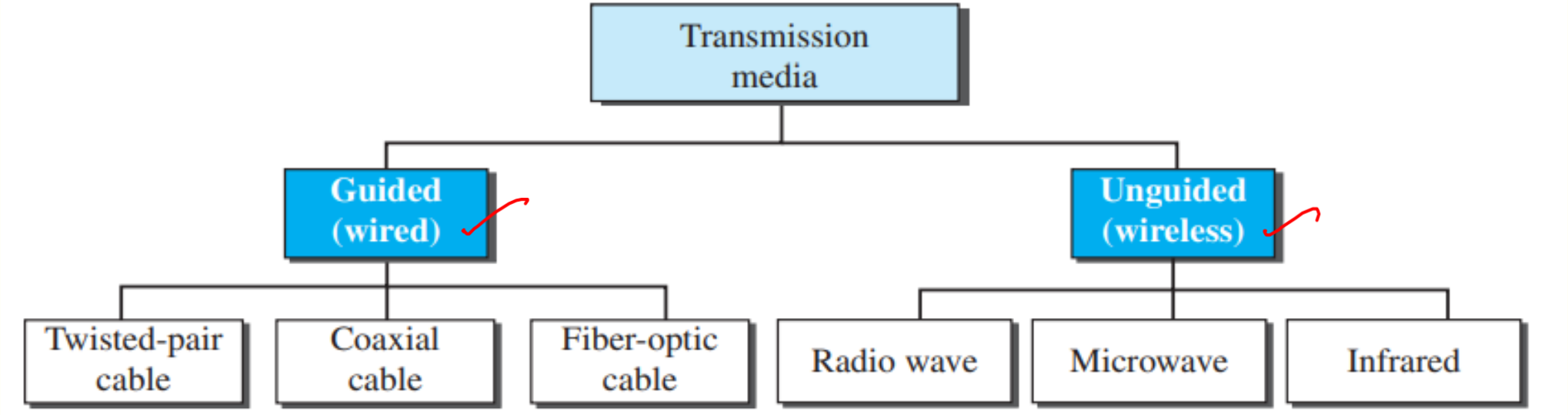


Contd…

- A transmission medium - anything that can carry information from a source to a destination.
- For example, the transmission medium for two people having a dinner conversation is the air.
- In data communications - free space, metallic cable, or fiber-optic cable.
- The information is usually a signal that is the result of a conversion of data from another form.
- Communication by telegraph was slow and dependent on a metallic medium

Contd…

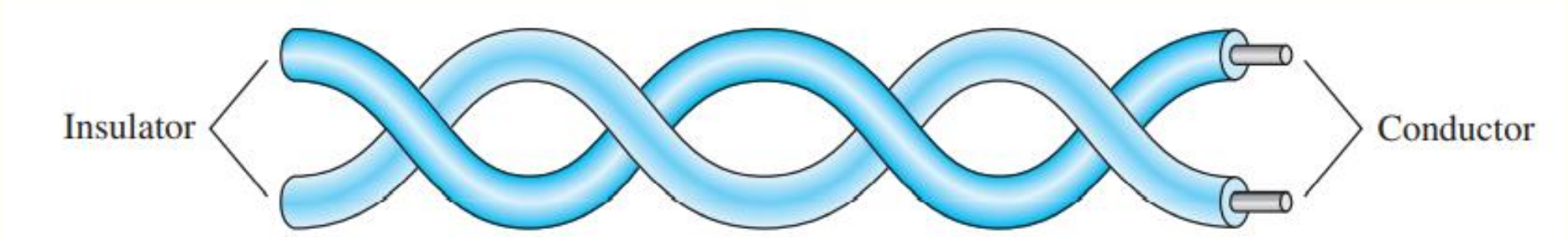
- In telecommunications, transmission media can be divided into two broad categories: guided and unguided.
- Guided media include twisted-pair cable, coaxial cable, and fiber-optic cable
- Unguided medium is free space



Guided Media

- Guided media - provide a channel from one device to another, include twisted-pair cable, coaxial cable, and fiber-optic cable.
- A signal traveling along any of these media is directed and contained by the physical limits of the medium.
- Twisted-pair and coaxial cable use metallic (copper) conductors that accept and transport signals in the form of electric current.
- Optical fiber is a cable that accepts and transports signals in the form of light.

Twisted-Pair Cable: A twisted pair consists of two conductors (normally copper), each with its own plastic insulation



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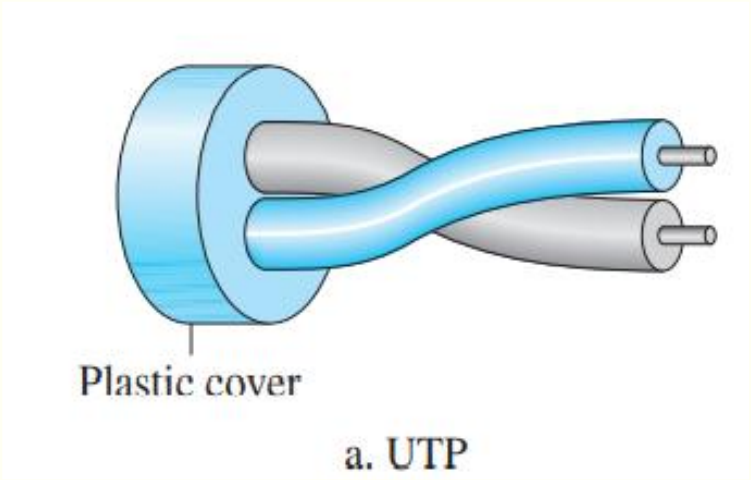
- One of the wires is used to carry signals to the receiver, and the other is used only as a ground reference.
- The receiver uses the difference between the two.
- In addition to the signal sent by the sender on one of the wires, interference (noise) and crosstalk may affect both wires and create unwanted signals.
- If the two wires are parallel, the effect of these unwanted signals is not the same in both wires because they are at different locations relative to the noise or crosstalk sources (e.g., one is closer and the other is farther).
- This results in a difference at the receiver.

Contd…

- By twisting the pairs, a balance is maintained.
- For example, suppose in one twist, one wire is closer to the noise source and the other is farther; in the next twist, the reverse is true.
- Twisting makes it probable that both wires are equally affected by external influences (noise or crosstalk).
- The receiver, which calculates the difference between the two, receives no unwanted signals.
- The unwanted signals are mostly canceled out. The number of twists per unit of length (e.g., inch) has some effect on the quality of the cable.

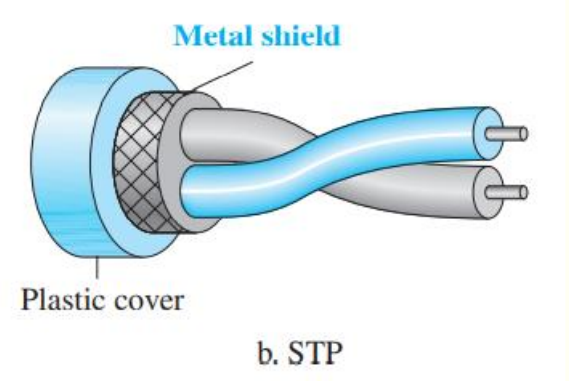
Unshielded Versus Shielded Twisted-Pair Cable

- The most common twisted-pair cable used in communications is referred to as unshielded twisted-pair (UTP).



a. UTP

- IBM has also produced a version of twisted-pair cable for its use, called shielded twisted-pair (STP)
- STP cable has a metal foil or braided mesh covering that encases each pair of insulated conductors.
- Improves the quality of cable by preventing the penetration of noise or crosstalk, it is bulkier and more expensive.



b. STP

Contd…

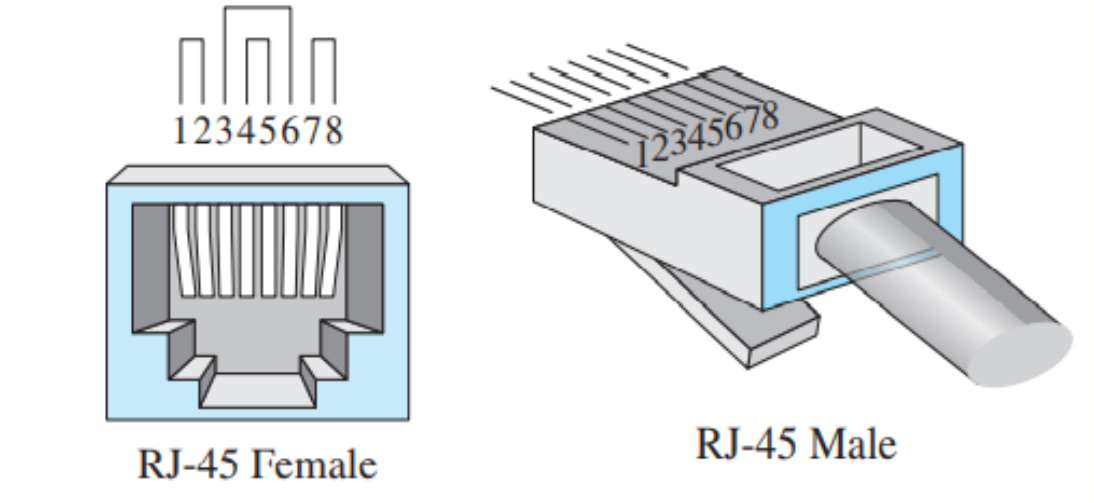
Category	*Specification*	*Data Rate (Mbps)*	*Use*
1	Unshielded twisted-pair used in telephone	< 0.1	Telephone
2	Unshielded twisted-pair originally used in T lines	2	T-1 lines
3	Improved CAT 2 used in LANs	10	LANs
4	Improved CAT 3 used in Token Ring networks	20	LANs
5	Cable wire is normally 24 AWG with a jacket and outside sheath	100	LANs

Categories of unshielded twisted-pair cables

Category	*Specification*	*Data Rate (Mbps)*	*Use*
5E	An extension to category 5 that includes extra features to minimize the crosstalk and electromagnetic interference	125	LANs
6	A new category with matched components coming from the same manufacturer. The cable must be tested at a 200-Mbps data rate.	200	LANs
7	Sometimes called *SSTP (shielded screen twisted-pair)*. Each pair is individually wrapped in a helical metallic foil followed by a metallic foil shield in addition to the outside sheath. The shield decreases the effect of crosstalk and increases the data rate.	600	LANs

Connectors

- The most common UTP connector is RJ45 (RJ stands for registered jack).
- The RJ45 is a keyed connector, meaning the connector can be inserted in only one way

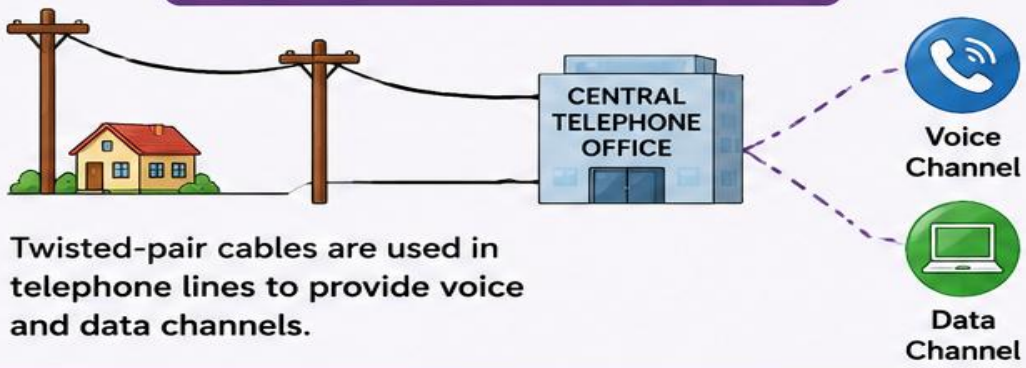


TWISTED-PAIR CABLES



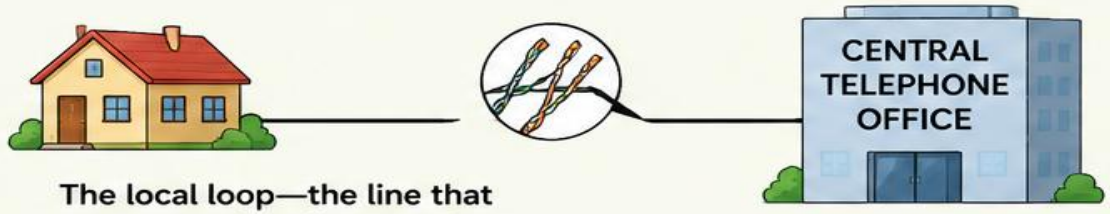
Twisted-pair cables consist of pairs of insulated copper wires twisted together to reduce electromagnetic interference (EMI).

1. USED IN TELEPHONE LINES



Twisted-pair cables are used in telephone lines to provide voice and data channels.

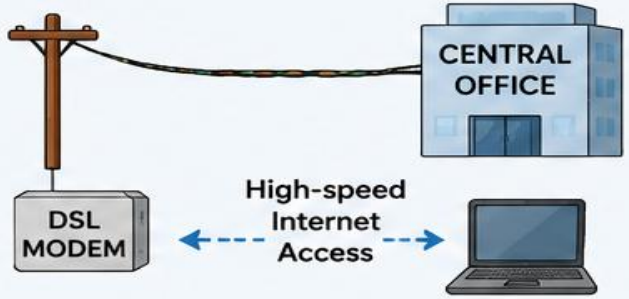
2. LOCAL LOOP (LAST MILE CONNECTION)



The local loop—the line that connects subscribers to the central telephone office—commonly consists of unshielded twisted-pair cables.

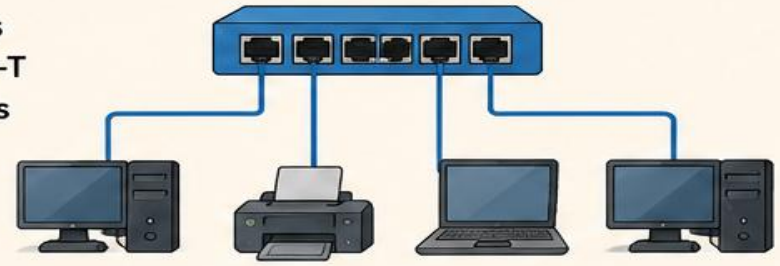
3. USL (DIGITAL SUBSCRIBER LINE)

DSL lines used by telephone companies to provide high-data-rate connections also use the high-bandwidth capability of unshielded twisted-pair cables.



4. USED IN LOCAL-AREA NETWORKS (LANs)

LAN standards such as 10Base-T and 100Base-T use twisted-pair cables to connect devices within a network.



10Base-T / 100Base-T

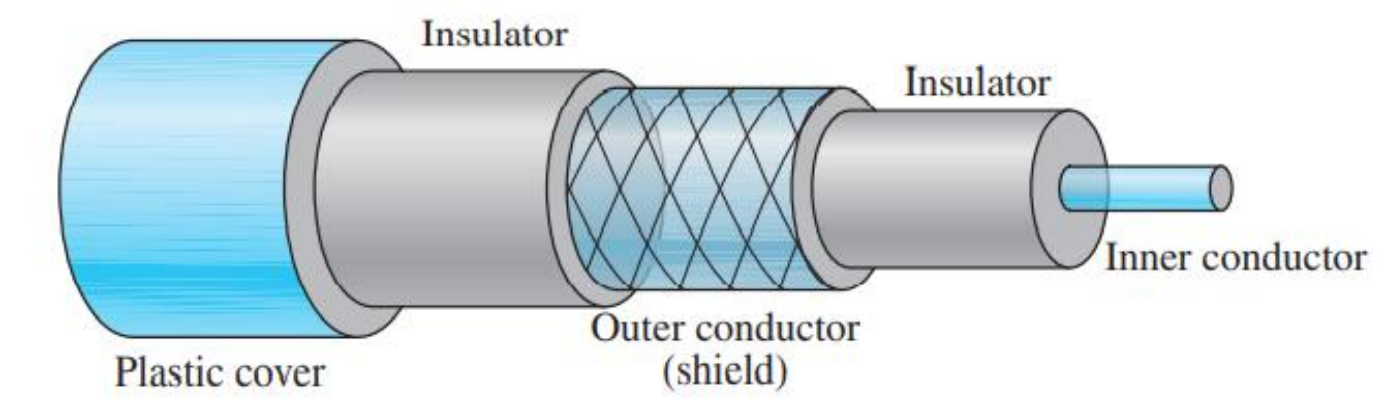
KEY POINTS

- Used for both voice and data transmission.
- Twisting reduces electromagnetic interference.
- Supports high-speed data services like DSL and LANs.
- Widely used in telecommunications and networking.

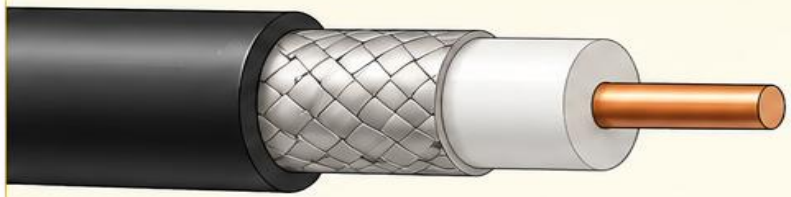
Applications

- Twisted-pair cables are used in telephone lines to provide voice and data channels.
- The local loop—the line that connects subscribers to the central telephone office— commonly consists of unshielded twisted-pair cables.
- The DSL lines that are used by the telephone companies to provide high-data-rate connections also use the high-bandwidth capability of unshielded twisted-pair cables.
- Local-area networks - 10Base-T and 100Base-T, also use twisted-pair cables

Coaxial Cable

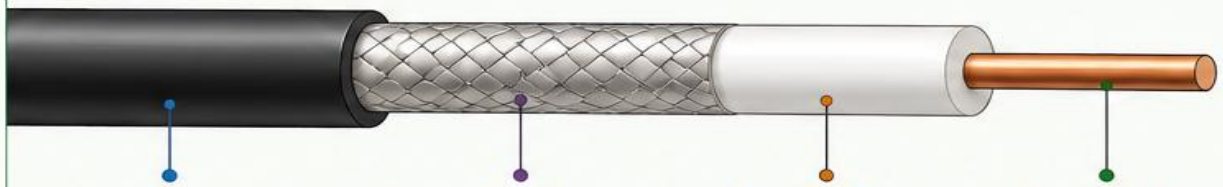


COAXIAL CABLE



- Coaxial cable (or coax) carries signals of higher frequency ranges than those in twisted pair cable.
- Instead of two wires, coax has a central core conductor of solid or stranded wire (usually copper).
- It is designed to provide better shielding and higher bandwidth.

CONSTRUCTION OF COAXIAL CABLE



Outer Plastic Cover
Protects the cable from physical damage and environment.

Outer Conductor (Shield)
Metal foil, braid, or both. Acts as a shield against noise and also as the second conductor.

Insulating Sheath (Dielectric)
Insulates the core conductor from the outer conductor.

Central Core Conductor
Solid or stranded copper wire that carries the signal.

KEY FUNCTIONS

Shield Against Noise
The outer metallic wrapping prevents electromagnetic interference (EMI) from affecting the signal.

Completes the Circuit
The outer conductor acts as the second conductor, providing the return path for the signal.

Higher Performance
Supports higher frequency signals and higher data transmission rates.

COAXIAL CABLE STANDARDS (RG RATINGS)

Coaxial cables are categorized by their Radio Government (RG) ratings.

Each RG number denotes a unique set of physical specifications, including:

Inner Conductor Wire Gauge



Thickness of the central copper wire.

Inner Insulator (Dielectric)



Thickness and type of insulating material.

Shield Construction



Type of shield (braid, foil, or both) and its construction.

Outer Casing



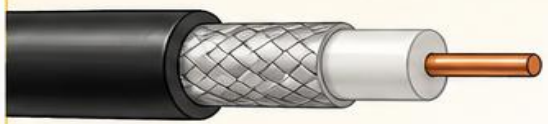
Size and type of the outer insulating sheath.

Overall Specifications



Determines the cable's performance and suitability for applications.

COAXIAL CABLE CONNECTORS



Used to connect coaxial cable to devices.

The most common type used today is the **Bayonet Neill-Concelman (BNC) connector.**

BNC connectors are easy to attach and remove with a twist-lock (bayonet) mechanism.

1. BNC CONNECTOR



Connects the end of the coaxial cable to a device, such as a TV set.

EXAMPLE USE

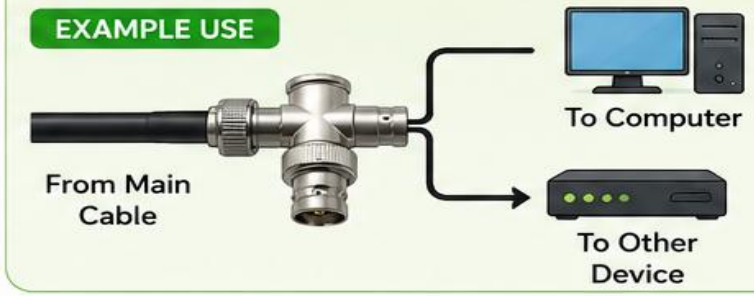


2. BNC T CONNECTOR



Used in Ethernet networks to branch out to a connection to a computer or other device.

EXAMPLE USE



3. BNC TERMINATOR



Used at the end of the cable to prevent the reflection of the signal.

EXAMPLE USE



Prevents signal reflection at the end of the cable.

QUICK SUMMARY

BNC Connector
Connects cable to a device.

BNC T Connector
Branches the connection in networks.

BNC Terminator
Ends the cable and prevents signal reflection.

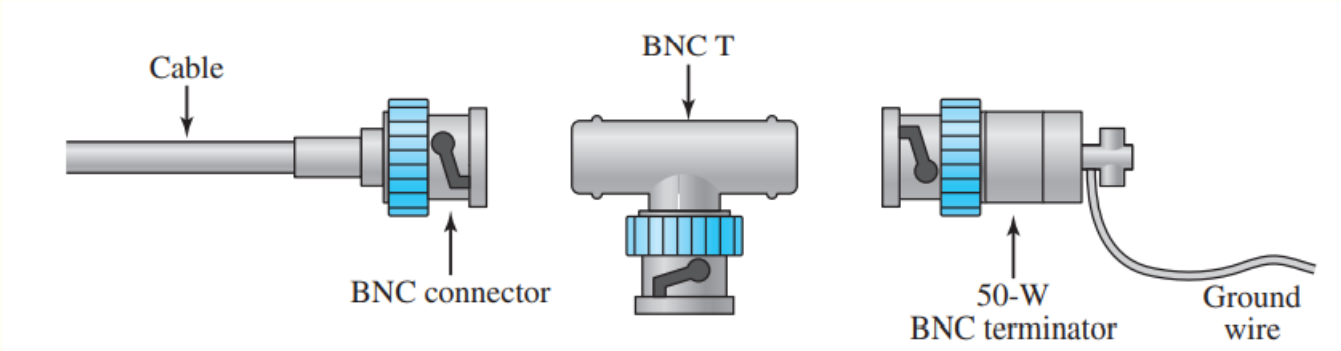
Remember:
Proper connectors ensure better signal quality and reliable communication.

Contd…

Categories of coaxial cables

Category	*Impedance*	*Use*
RG-59	75 Ω	Cable TV
RG-58	50 Ω	Thin Ethernet
RG-11	50 Ω	Thick Ethernet

BNC connectors



Coaxial Cable Connectors

- To connect coaxial cable to devices - coaxial connectors.
- The most common type of connector used today is the Bayonet Neill-Concelman (BNC) connector

Three popular types of these connectors: BNC connector, BNC T connector, and BNC terminator

- The BNC connector is used to connect the end of the cable to a device, such as a TV set.
- The BNC T connector is used in Ethernet networks to branch out to a connection to a computer or other device.
- The BNC terminator is used at the end of the cable to prevent the reflection of the signal.

APPLICATIONS – COAXIAL CABLE

Coaxial cable is used in many communication and TV networks.

1. Uses of Coaxial Cable



Analog Telephone Networks

A single coaxial network could carry up to **10,000** voice signals.



Digital Telephone Networks

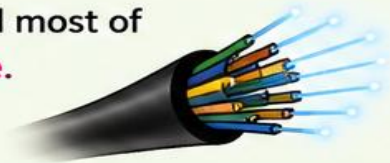
A single coaxial cable could carry digital data up to **600 Mbps**.



Cable TV Network

The entire network (TV channels, video and data) used coaxial cable.

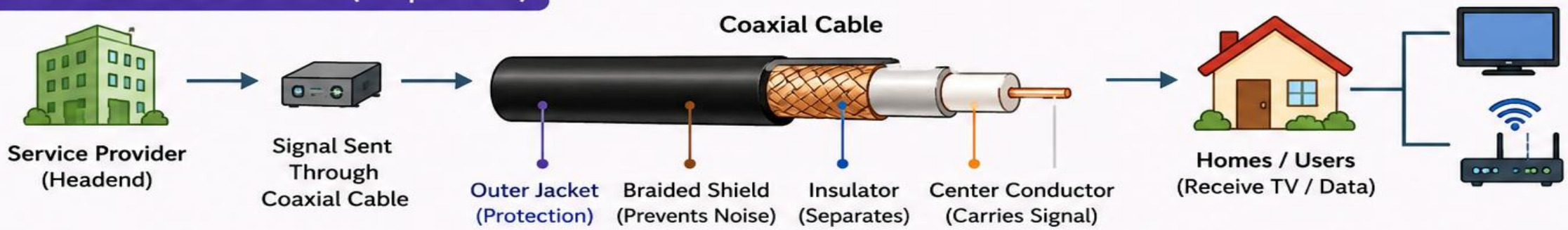
2 Later, cable TV providers replaced most of the media with **fiber-optic cable**.



3 Cable TV uses **RG-59** coaxial cable.



How Coaxial Cable Works (Simple View)



Key idea: Coaxial cable is strong, reliable, and used for voice, data, and TV services over long distances.

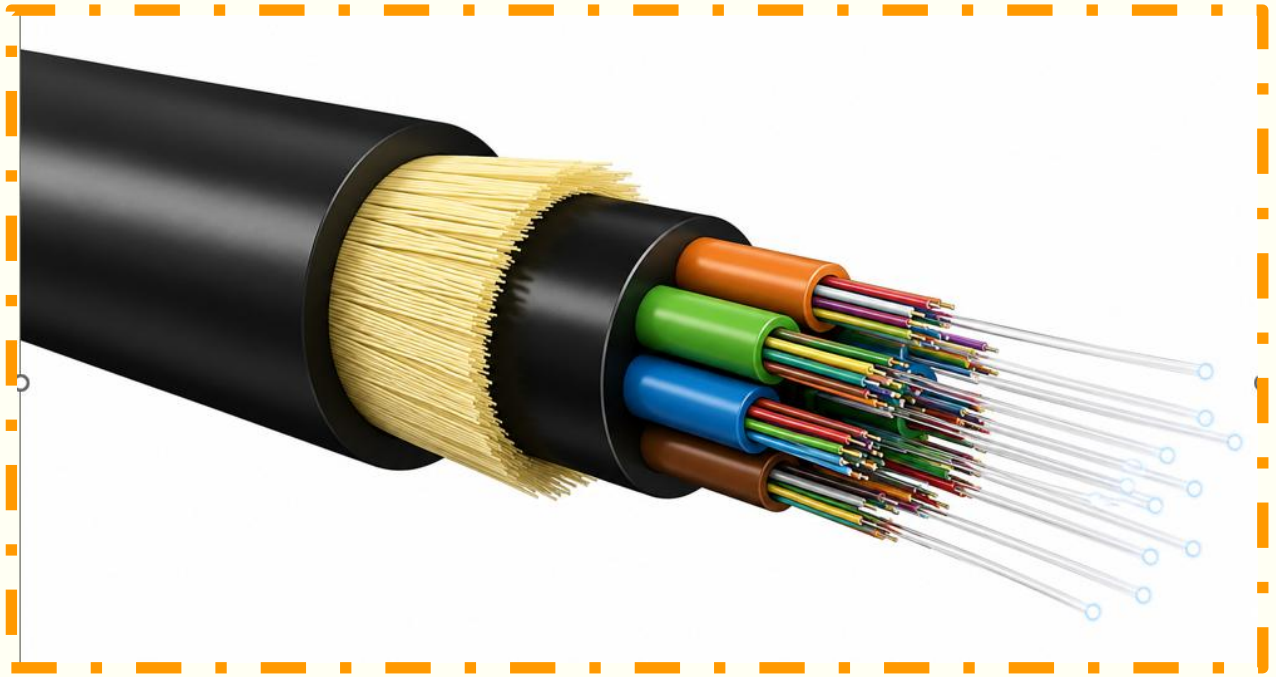
Applications

- Coaxial cable:
  - Analog telephone networks where a single coaxial network could carry 10,000 voice signals.
  - Digital telephone networks where a single coaxial cable could carry digital data up to 600 Mbps.
  - Cable TV network, the entire network used coaxial cable.
- Later, cable TV providers replaced most of the media with fiber-optic cable;
- Cable TV uses RG-59 coaxial cable.

Contd…

- Coaxial cable is in traditional Ethernet LANs
- Because of its high bandwidth, and consequently high data rate, coaxial cable was chosen for digital transmission in early Ethernet LANs.
- The 10Base-2, or Thin Ethernet, uses RG-58 coaxial cable with BNC connectors to transmit data at 10 Mbps with a range of 185 m.
- The 10Base5, or Thick Ethernet, uses RG-11 (thick coaxial cable) to transmit 10 Mbps with a range of 5000 m.
- Thick Ethernet has specialized connectors.

Fiber-Optic Cable



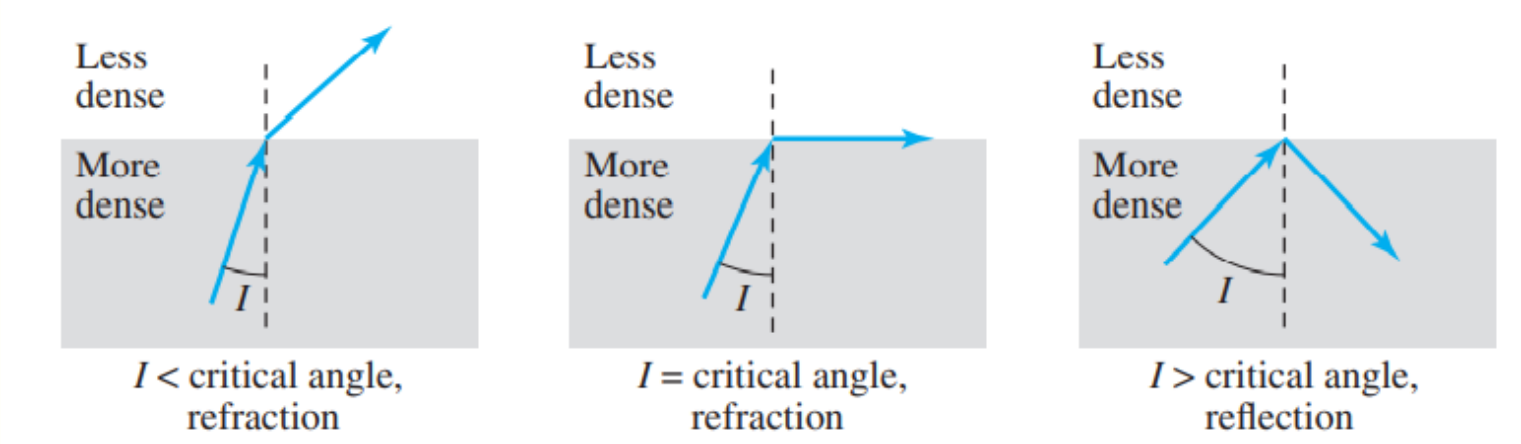
Fiber-Optic Cable

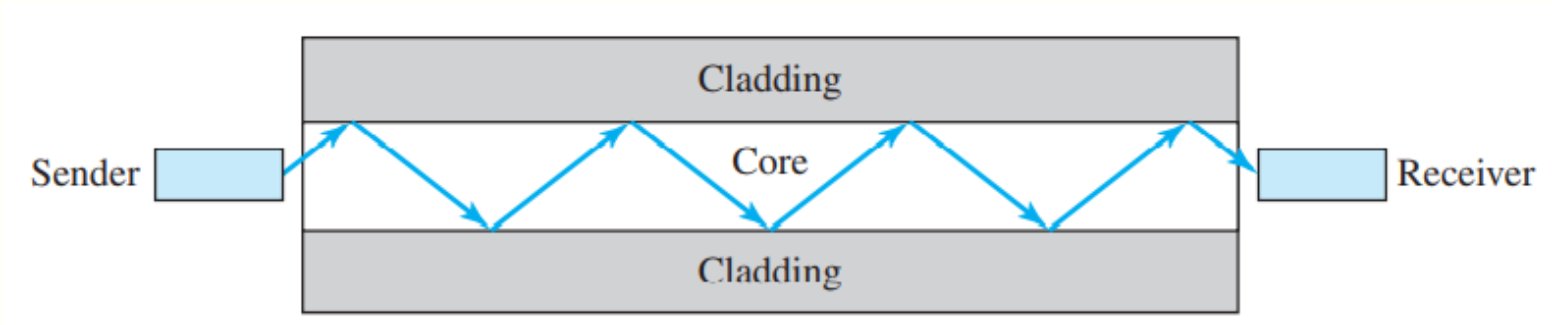
- A fiber-optic cable is made of glass or plastic and transmits signals in the form of light.
- Light travels in a straight line as long as it is moving through a single uniform substance. If a ray of light traveling through one substance suddenly enters another substance (of a different density), the ray changes direction.
- if the angle of incidence I (the angle the ray makes with the line perpendicular to the interface between the two substances) is less than the critical angle, the ray refracts and moves closer to the surface.
- If the angle of incidence is equal to the critical angle, the light bends along the interface.
- If the angle is greater than the critical angle, the ray reflects (makes a turn) and travels again in the denser.

Contd…

- Optical fibers use reflection to guide light through a channel.
- A glass or plastic core is surrounded by a cladding of less dense glass or plastic.
- The difference in density of the two materials must be such that a beam of light moving through the core is reflected off the cladding instead of being refracted into it.

Bending of light ray & Optical fiber

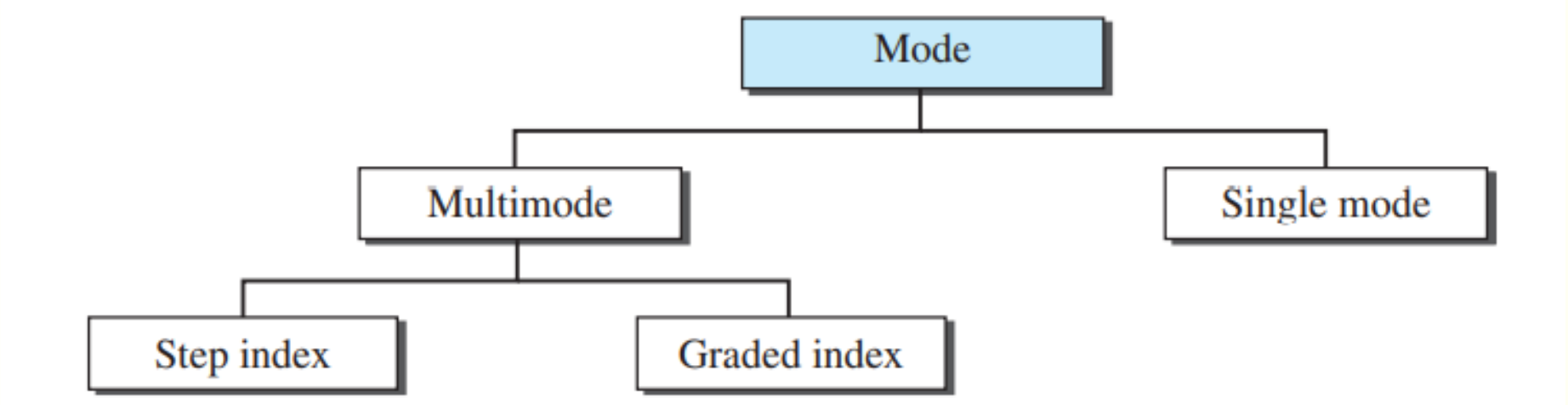




https://www.youtube.com/watch?v=jZOg39v73c4

Propagation Modes

- Current technology supports two modes (multimode and single mode) for propagating light along optical channels, each requiring fiber with different physical characteristics.
- Multimode can be implemented in two forms: step-index or graded-index



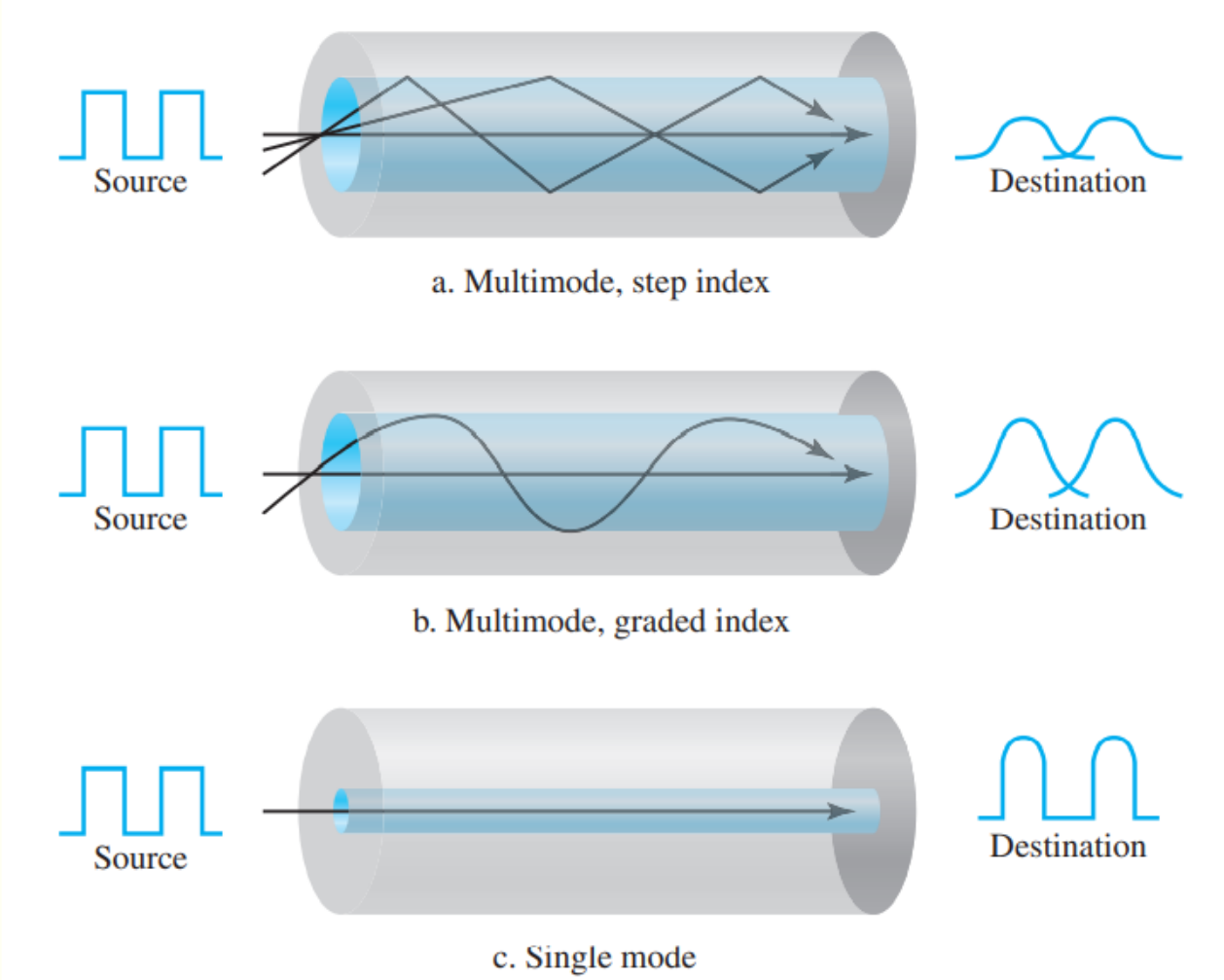
Multimode

- Multimode - multiple beams from a light source move through the core in different paths.
- In multimode step-index fiber, the density of the core remains constant from the center to the edges.
- A beam of light moves through this constant density in a straight line until it reaches the interface of the core and the cladding.
- At the interface, there is an abrupt change due to a lower density; this alters the angle of the beam's motion.
- The term step-index refers to the suddenness of this change, which contributes to the distortion of the signal as it passes through the fiber.

Contd…

- A second type of fiber - multimode graded-index fiber, decreases this distortion of the signal through the cable.
- The index of refraction is related to density.
- A graded index fiber is one with varying densities.
- Density is highest at the center of the core and decreases gradually to its lowest at the edge.

Contd…


Source
Destination
a. Multimode, step index
Source
Destination
b. Multimode, graded index
Source
Destination
c. Single mode

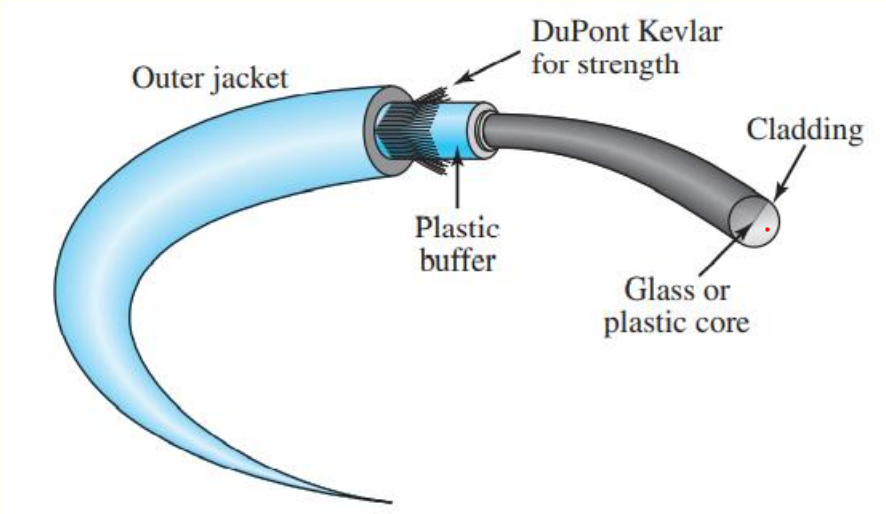
Single-Mode

- Single-mode uses step-index fiber and a highly focused source of light that limits beams to a small range of angles, all close to the horizontal.
- The decrease in density results in a critical angle that is close enough to 90° to make the propagation of beams almost horizontal.
- Propagation of different beams is almost identical, and delays are negligible.
- All the beams arrive at the destination “together” and can be recombined with little distortion to the signal.

Fiber types

Type	*Core (μm)*	*Cladding (μm)*	*Mode*
50/125	50.0	125	Multimode, graded index
62.5/125	62.5	125	Multimode, graded index
100/125	100.0	125	Multimode, graded index
7/125	7.0	125	Single mode

Fiber construction

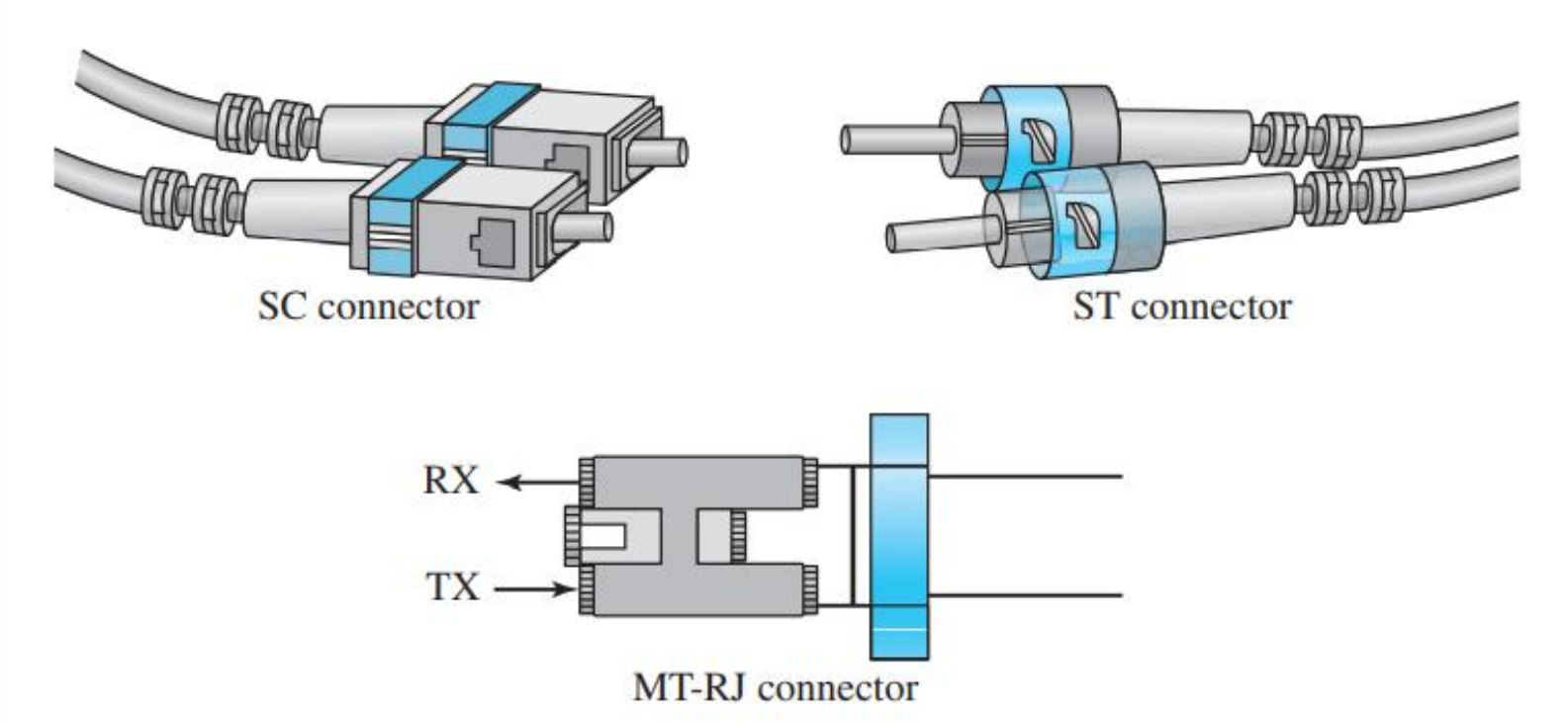


Fiber-Optic Cable Connectors

There are three types of connectors for fiber-optic cables.

- The subscriber channel (SC) connector is used for cable TV.
  - It uses a push/pull locking system.
- The straight-tip (ST) connector is used for connecting cable to networking devices.
  - It uses a bayonet locking system and is more reliable than SC.
- MT-RJ is a connector that is the same size as RJ45.

Fiber-optic cable connectors



Applications

- Fiber-optic cable is often found in backbone networks because its wide bandwidth is cost-effective.
- wavelength-division multiplexing (WDM) - data at a rate of 1600 Gbps.
- The SONET network.
- Local-area networks such as 100Base-FX network (Fast Ethernet) and 1000Base-X also use fiber-optic cable.

Advantages

- Fiber-optic cable has several advantages over metallic cable (twisted-pair or coaxial).
- Higher bandwidth - Fiber-optic cable can support dramatically higher bandwidths (and hence data rates) than either twisted-pair or coaxial cable.
- Currently, data rates and bandwidth utilization over fiber-optic cable are limited not by the medium but by the signal generation and reception technology available.
- Less signal attenuation - Fiber-optic transmission distance is significantly greater than that of other guided media. A signal can run for 50 km without requiring regeneration. We need repeaters every 5 km for coaxial or twisted-pair cable.

Contd…

- Immunity to electromagnetic interference. Electromagnetic noise cannot affect fiber-optic cables.
- Resistance to corrosive materials. Glass is more resistant to corrosive materials than copper.
- Light weight. Fiber-optic cables are much lighter than copper cables.
- Greater immunity to tapping. Fiber-optic cables are more immune to tapping than copper cables. Copper cables create antenna effects that can easily be tapped.

Disadvantages

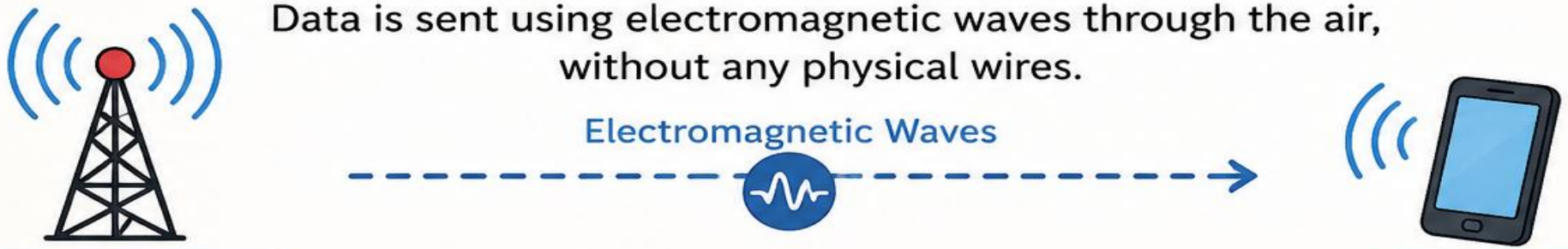
There are some disadvantages in the use of optical fiber.

- **Installation and maintenance**. Fiber-optic cable is a relatively new technology. Its installation and maintenance require expertise that is not yet available everywhere.
- **Unidirectional light propagation**. Propagation of light is unidirectional. If we need bidirectional communication, two fibers are needed.
- **Cost.** The cable and the interfaces are relatively more expensive than those of other guided media. If the demand for bandwidth is not high, often the use of optical fiber cannot be justified.

UNGUIDED MEDIA

UNGUIDED MEDIA: WIRELESS

Data is sent using electromagnetic waves through the air, without any physical wires.



Wireless communication uses electromagnetic waves in the **range of 3 kHz to 900 THz.**

EXAMPLES



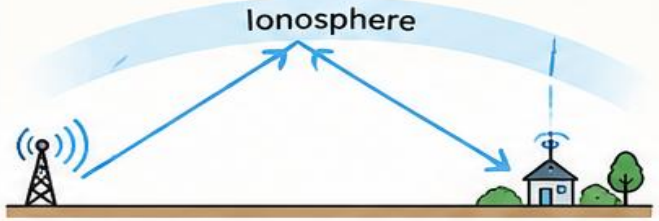
HOW WIRELESS SIGNALS TRAVEL

1. GROUND PROPAGATION



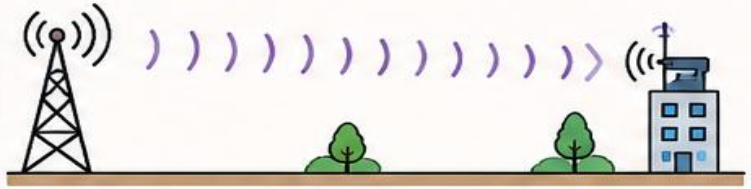
Follows the Earth's surface.

2. SKY PROPAGATION



Waves bounce off the ionosphere and travel long distances.

3. LINE-OF-SIGHT PROPAGATION



Waves travel in a straight path between the transmitter and receiver.

KEY POINT

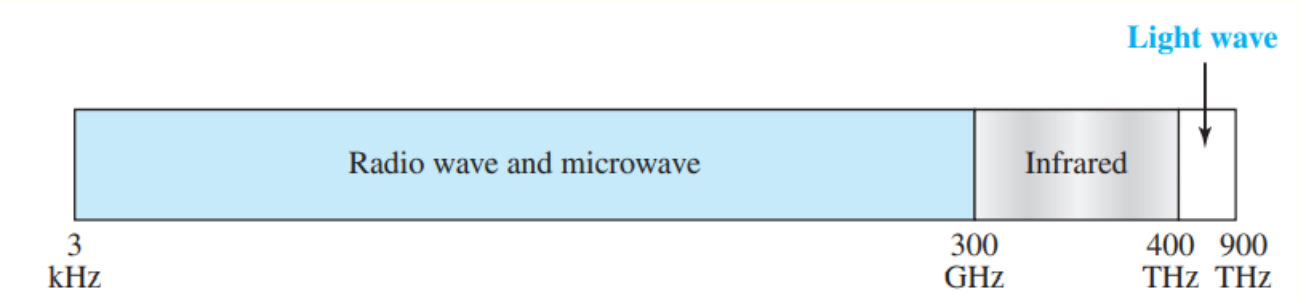
Wireless signals travel through space and can be received by anyone with a compatible device.



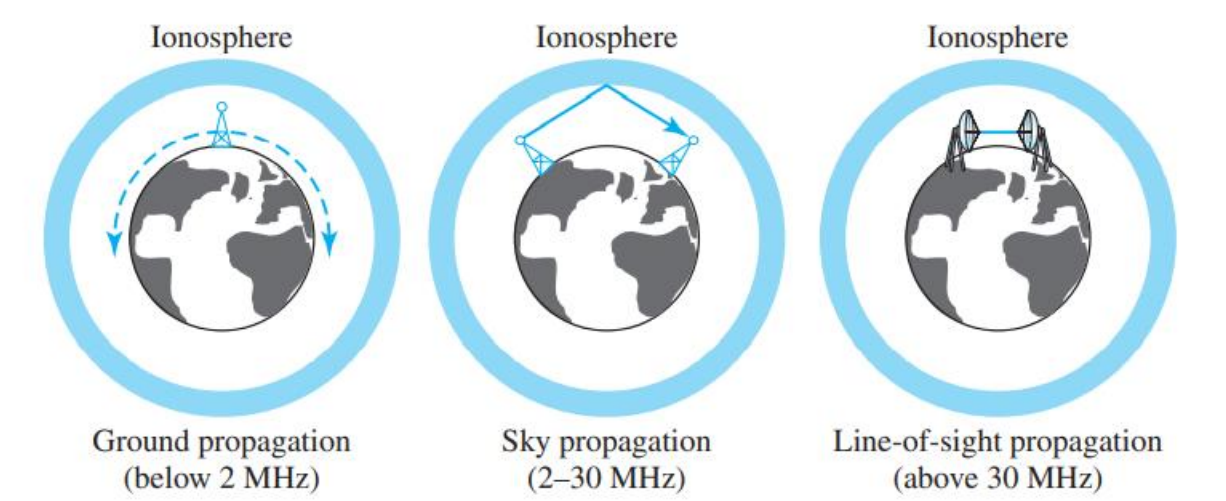
UNGUIDED MEDIA: WIRELESS

- Unguided medium transport electromagnetic waves without using a physical conductor.
- This type of communication is often referred to as wireless communication.
- Signals are normally broadcast through free space and thus are available to anyone who has a device capable of receiving them.
- The part of the electromagnetic spectrum, ranging from 3 kHz to 900 THz, used for wireless communication.
- Unguided signals can travel from the source to the destination in several ways: ground propagation, sky propagation, and line-of-sight propagation

Electromagnetic spectrum for wireless communication



Propagation methods



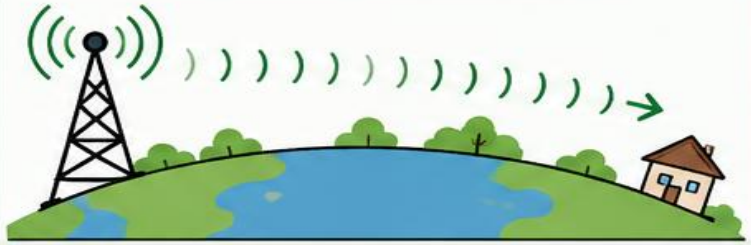
UNGUIDED MEDIA: WIRELESS

Wireless communication uses **electromagnetic waves.**
Signals travel through air/space **without any physical wires.**

Radio | TV | Wi-Fi | Mobile

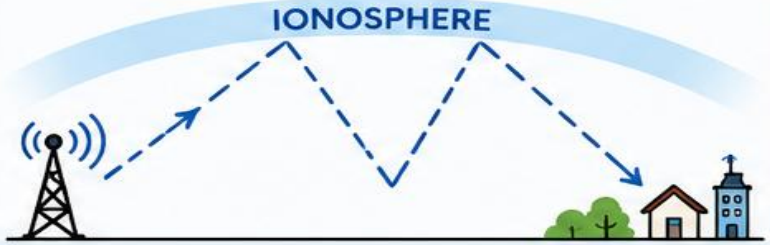
HOW RADIO WAVES TRAVEL

1. GROUND PROPAGATION



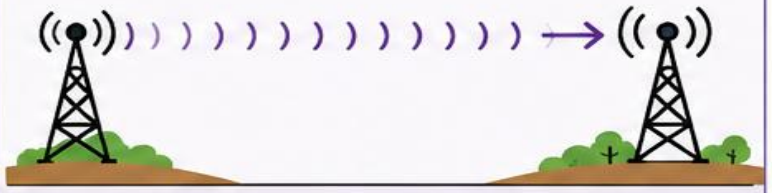
- Low frequency waves follow the curvature of the Earth.
- Travel through the lowest part of the atmosphere.
- Radiate in all directions.
- More power = More distance.

2. SKY PROPAGATION



- Higher frequency waves go up to the ionosphere.
- They get reflected back to Earth.
- Allows long distance communication with less power.

3. LINE-OF-SIGHT PROPAGATION



- Very high frequency waves travel in straight lines from one antenna to another.
- Antennas must face each other.
- Need height or short distance to avoid Earth's curvature.
- Signals cannot be perfectly focused.

RADIO WAVES & MICROWAVES BANDS

The radio & microwave part of the electromagnetic spectrum is divided into 8 bands (regulated by government authorities).

	1	2	3	4	5	6	7	8
	VLF	LF	MF	HF	VHF	UHF	SHF	EHF
	Very Low Frequency	Low Frequency	Medium Frequency	High Frequency	Very High Frequency	Ultra High Frequency	Super High Frequency	Extremely High Frequency
	3 kHz – 30 kHz	30 kHz – 300 kHz	300 kHz – 3 MHz	3 MHz – 30 MHz	30 MHz – 300 MHz	300 MHz – 3 GHz	3 GHz – 30 GHz	30 GHz – 300 GHz

Lower frequencies = longer distance, less data | Higher frequencies = shorter distance, more data

Bands

Band	*Range*	*Propagation*	*Application*
very low frequency (VLF)	3–30 kHz	Ground	Long-range radio navigation
low frequency (LF)	30–300 kHz	Ground	Radio beacons and navigational locators

Band	*Range*	*Propagation*	*Application*
middle frequency (MF)	300 kHz–3 MHz	Sky	AM radio
high frequency (HF)	3–30 MHz	Sky	Citizens band (CB), ship/aircraft
very high frequency (VHF)	30–300 MHz	Sky and line-of-sight	VHF TV, FM radio
ultrahigh frequency (UHF)	300 MHz–3 GHz	Line-of-sight	UHF TV, cellular phones, paging, satellite
superhigh frequency (SF)	3–30 GHz	Line-of-sight	Satellite
extremely high frequency (EHF)	30–300 GHz	Line-of-sight	Radar, satellite

Contd…

- In ground propagation, radio waves travel through the lowest portion of the atmosphere, hugging the earth.
- These low-frequency signals emanate in all directions from the transmitting antenna and follow the curvature of the planet.
- Distance depends on the amount of power in the signal: The greater the power, the greater the distance.
- In sky propagation, higher-frequency radio waves radiate upward into the ionosphere (the layer of atmosphere where particles exist as ions) where they are reflected back to earth.
- This type of transmission allows for greater distances with lower output power.

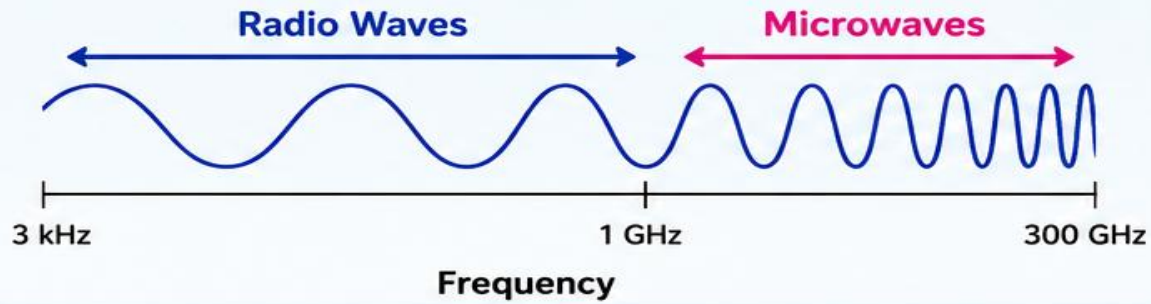
Contd…

- In line-of-sight propagation, very high-frequency signals are transmitted in straight lines directly from antenna to antenna.
- Antennas must be directional, facing each other, and either tall enough or close enough together not to be affected by the curvature of the earth.
- Line-of-sight propagation is tricky because radio transmissions cannot be completely focused.
- The section of the electromagnetic spectrum defined as radio waves and microwaves is divided into eight ranges, called bands, each regulated by government authorities.
- These bands are rated from very low frequency (VLF) to extremely high frequency (EHF).

Radio Waves

1. What are Radio Waves?

Radio waves and microwaves are electromagnetic waves.





Radio waves:
Frequencies between
3 kHz and 1 GHz.



Microwaves:
Frequencies between
1 GHz and **300 GHz.**

2. Radio Waves are Omnidirectional

They spread out equally in all directions from the antenna.



3. Propagation in All Directions

When an antenna transmits radio waves, they are propagated in all directions.



4. Susceptible to Interference

Radio waves from one antenna can be disturbed by another antenna using the same frequency or band.



Example


FM Radio


Television


Satellite Communication


Wi-Fi


Bluetooth


Mobile Communication

Key Idea

Radio waves: **3 kHz – 1 GHz**
Microwaves: **1 GHz – 300 GHz**
Spread in all directions and can be interfered by other signals.

Radio Waves

- Radio waves and microwaves, electromagnetic waves ranging in frequencies between 3 kHz and 1 GHz are normally called radio waves; waves ranging in frequencies between 1 and 300 GHz are called microwaves.
- Radio waves - omnidirectional.
- When an antenna transmits radio waves, they are propagated in all directions.
- The radio waves transmitted by one antenna are susceptible to interference by another antenna that may send signals using the same frequency or band.

Contd…

- Radio waves, particularly those waves that propagate in the sky mode, can travel long distances.
- The radio wave band is relatively narrow, just under 1 GHz, compared to the microwave band.
- When this band is divided into subbands, the subbands are also narrow, leading to a low data rate for digital communications

Omnidirectional Antenna

- Radio waves use omnidirectional antennas that send out signals in all directions. Based on the wavelength, strength, and the purpose of transmission, we can have several types of antennas
- The omnidirectional characteristics of radio waves make them useful for multicasting, in which there is one sender but many receivers. AM and FM radio, television, maritime radio, cordless phones, and paging are examples of multicasting



MICROWAVES

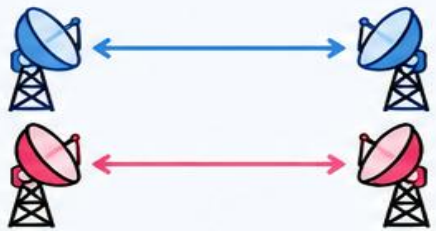
Electromagnetic waves having frequencies between 1 and 300 GHz.

1 Unidirectional



Microwaves travel in one direction. Sending and receiving antennas must be aligned.

2 No Interference



One pair of aligned antennas does not interfere with another pair.

3 Line-of-Sight



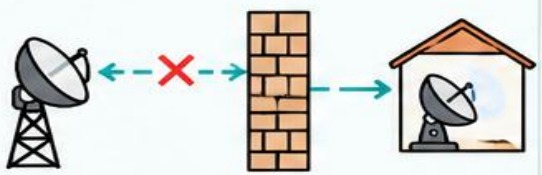
Microwave signals travel in a straight line. Towers must be in direct sight.

4 Need Tall Towers / Repeaters



Earth's curvature and obstacles block the signal. Repeaters or very tall towers are needed for long distance.

5 Does Not Penetrate Walls



Microwaves cannot pass through walls. Receivers inside buildings may not get the signal.

6 Wide Band



~299 GHz

Microwave band is wide, so wider subbands can be assigned. High data rate is possible.

7 Permission Required



Using certain parts of the microwave band requires permission from authorities.

Applications



Satellite Communication | Telecom Backhaul | Wireless Networks

Key Idea: Microwaves are fast, directional signals used for communication when antennas are aligned and in direct sight.

Microwaves

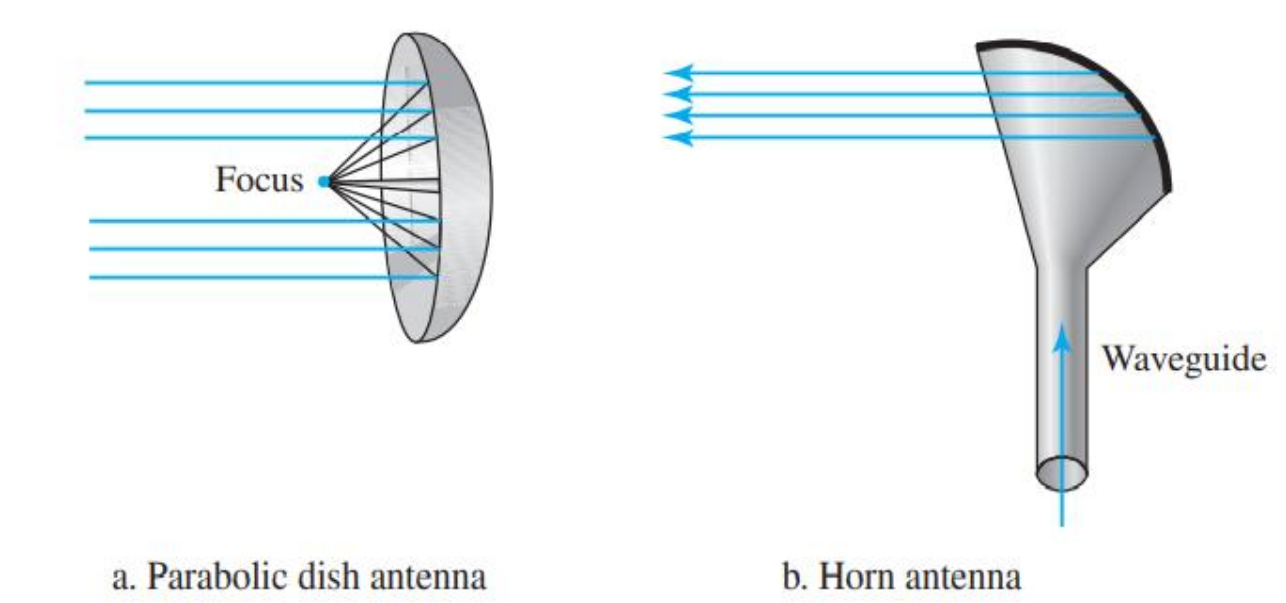
- Electromagnetic waves having frequencies between 1 and 300 GHz are called microwaves.
- Microwaves are unidirectional. The sending and receiving antennas need to be aligned.
- A pair of antennas can be aligned without interfering with another pair of aligned antennas

Characteristics of microwave propagation

- Microwave propagation is line-of-sight. Since the towers with the mounted antennas need to be in direct sight of each other, towers that are far apart need to be very tall.
- The curvature of the earth as well as other blocking obstacles do not allow two short towers to communicate by using microwaves. Repeaters are often needed for longdistance communication.
- Very high-frequency microwaves cannot penetrate walls. This characteristic can be a disadvantage if receivers are inside buildings.
- The microwave band is relatively wide, almost 299 GHz. Therefore wider subbands can be assigned, and a high data rate is possible.
- Use of certain portions of the band requires permission from authorities.

Unidirectional Antenna

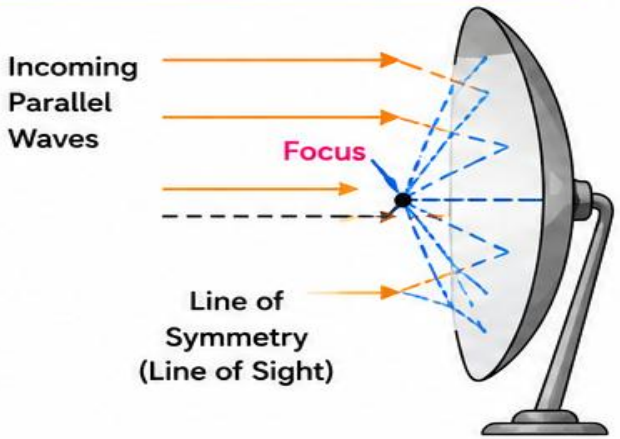
- Microwaves need unidirectional antennas that send out signals in one direction.
- Two types of antennas are used for microwave communications: the parabolic dish and the horn.



PARABOLIC DISH & HORN ANTENNA

A parabolic dish antenna uses the shape of a parabola to collect or send microwaves efficiently.

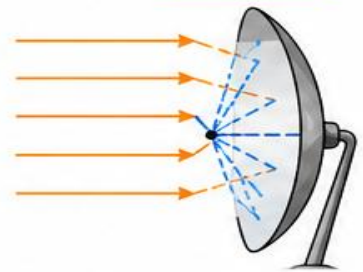
1. How Parabolic Dish Works



- Every line parallel to the line of symmetry reflects off the curve and intersects at the focus.
- The dish catches a wide range of waves and directs them to the focus.
- More signal is recovered at the focus than with a single-point receiver.

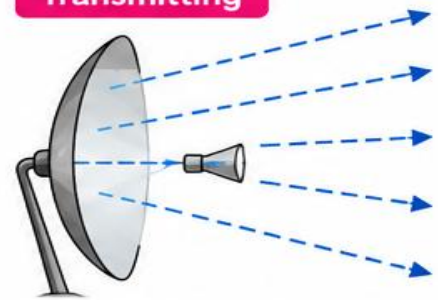
2. Sending and Receiving with Parabolic Dish

Receiving



Incoming microwaves hit the dish and are reflected to the feed (focus).

Transmitting

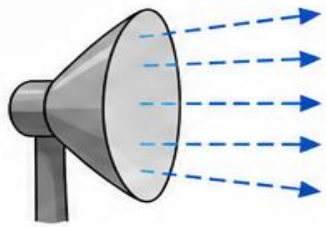


Outgoing signals from the horn hit the dish and are deflected outward.

3. Horn Antenna (Giant Scoop)

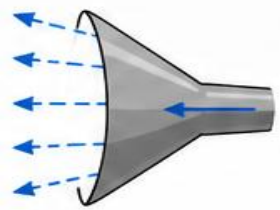
Transmit

Signals travel up the stem and the curved head sends them out in narrow, parallel beams.



Receive

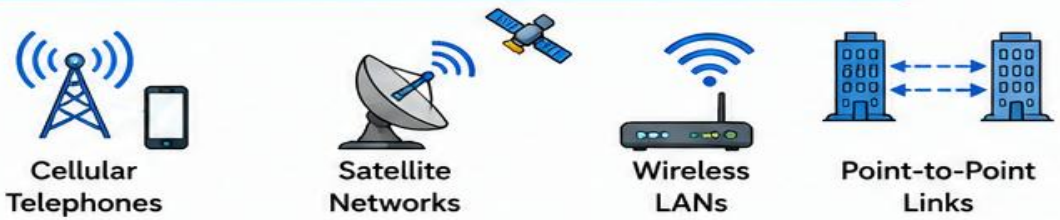
Incoming waves are collected by the scooped shape and directed down into the stem.



Parabolic Dish vs Horn Antenna

	Parabolic Dish	Horn Antenna
	Focuses waves to a point (the focus)	Collects or sends waves in a scooped shape
	Works like a funnel	Looks like a giant scoop
	High gain, narrow beam	High gain, narrow beam
	Used for both sending and receiving	Used for both sending and receiving

4. Where Microwaves are Used (Unicast Communication)



Key Idea

Parabolic dishes and horn antennas help in efficient sending and receiving of microwaves by focusing or directing signals in a specific direction.



Contd…

- A parabolic dish antenna is based on the geometry of a parabola: Every line parallel to the line of symmetry (line of sight) reflects off the curve at angles such that all the lines intersect in a common point called the focus.
- The parabolic dish works as a funnel, catching a wide range of waves and directing them to a common point.
- In this way, more of the signal is recovered than would be possible with a single-point receiver.
- Outgoing transmissions are broadcast through a horn aimed at the dish.
- The microwaves hit the dish and are deflected outward in a reversal of the receipt path.

Contd…

- A horn antenna looks like a gigantic scoop.
- Outgoing transmissions are broadcast up a stem (resembling a handle) and deflected outward in a series of narrow parallel beams by the curved head.
- Received transmissions are collected by the scooped shape of the horn, in a manner similar to the parabolic dish, and are deflected down into the stem.
- Microwaves are used for unicast communication such as cellular telephones, satellite networks, and wireless LANs.

INFRARED WAVES

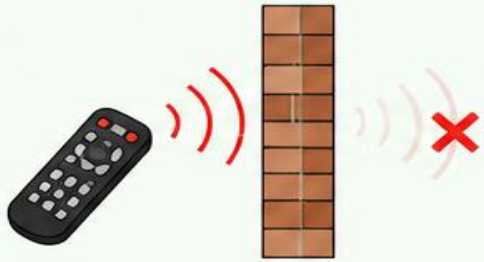
Infrared waves, with frequencies from 300 GHz to 400 THz
(wavelengths from 1 mm to 770 nm), can be used for short-range communication.

1. Short-Range Communication



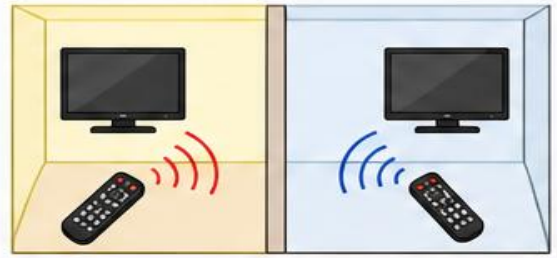
Infrared signals can be used for short-range communication in a closed area using line-of-sight propagation.

2. Cannot Penetrate Walls



Infrared waves, having high frequencies, cannot penetrate walls.

3. No Interference Between Systems



This prevents interference between one system and another; a system in one room cannot be affected by another system in the next room.

4. Line-of-Sight Propagation



Infrared signals require a direct line of sight between the transmitter and receiver.

Examples



TV Remote Control



IR Wireless Headphones



IR Data Transfer (Between Devices)



Motion Detectors

Infrared Spectrum



300 GHz — Infrared Region — 400 THz

(Wavelengths: 1 mm to 770 nm)

Key Idea: Infrared waves are ideal for short-range, line-of-sight communication as they cannot pass through walls and do not cause interference.